

Welcome to the Desert of the Real

esys app
Overview and Features Walkthrough

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Plan

- Intro
- Market Conditions
- HV Interface: Calculation of Historical Volatility
- IV Interface: Calculation of Implied Volatility
- VC Interface: Calculation of The Volatility Cone
- Conclusion
- References



Intro

1st impl (emodule/emod)

- Developer(s) Denis Khorkin
- Initial release July 6, 2016
- Written in VBA
- OS Microsoft Windows



VBA, which stands for Visual Basic for Applications, is a programming language developed by Microsoft. Excel, along with the other members of Microsoft Office, includes the VBA language (at no extra charge). In a nutshell, VBA is the tool that people use to develop programs that control Excel. VBA is a programming language, but it also serves as a macro language. In this context, a macro is a set of instructions Excel performs to imitate keystrokes and mouse actions.[1]

2nd impl (esys)

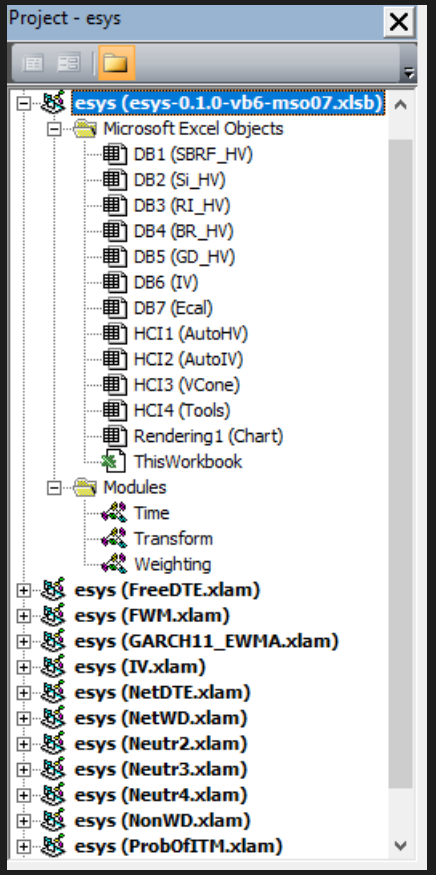
- Developer(s) Denis Khorkin
- Initial release Unreleased
- Repo https://gitflic.ru/project/0mega3/esys
- Written in Haskell
- OS Linux
- License BSD-3-Clause



Haskell is a general purpose, purely functional programming language incorporating many recent innovations in programming language design. Haskell provides higher-order functions, non-strict semantics, static polymorphic typing, user-defined algebraic datatypes, pattern-matching, list comprehensions, a module system, a monadic I/O system, and a rich set of primitive datatypes, including lists, arrays, arbitrary and fixed precision integers, and floating-point numbers.[2]



Proto-Architecture



Class Modules

- SBRF_HV .. GD_HV: Databases of Settlement Price, Historical Volatility (HV) for 100, 50, 20, 10 days for underlying assets;
- IV: Implied Volatility (IV) database;
- Ecal: Options Expiration Calendar;
- AutoHV, AutoIV: HCI for automatic calculation of HV and IV;
- VCone: HCI for automatic calculation of the Volatility Cone;
- Tools: Tools for working with futures and options;
- Chart: Rendering of 2d charts;
- ThisWorkbook: Refers to the workbook containing the currently running code.

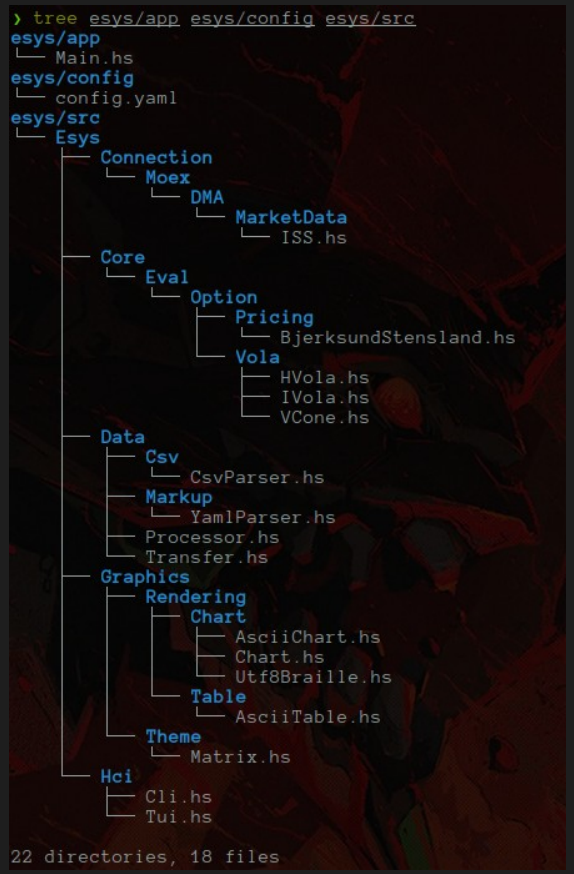
Modules

- Time: Working with time;
- Transform: Data transformation;
- Weighting: Data weighting.

Add-ins

- FWM: Fishburne Weighting Method;
- FreeDTE, NetDTE, NetWD, NonWD: HV normalizers;
- Neutr2, Neutr3, Neutr4: Risk neutralizers for 2, 3, 4 options;
- GARCH11_EWMA, IV, ProbOfITM: Not covered (for real-time data).

Architecture



- Main.hs: Main module to compile a standalone executable;
- config.yaml: The main configuration file;
- ISS.hs: The Moscow Exchange ISS information interface;
- BjerksundStensland.hs: The Bjerksund-Stensland American option model;
- HVola.hs: Historical volatility (HV) is calculated;
- IVola.hs: The agg value of implied volatility (IV) is calculated;
- VCone.hs: The volatility cone (VC) matrix is created;
- CsvParser.hs: CSV parsing and encoding;
- YamlParser.hs: YAML parsing;
- Processor.hs: Data processing (DP);
- Transfer.hs: Transferring data between the app and the disk;
- AsciiChart.hs: ASCII line charts in terminal ╭(･ㅂ･)╯;
- Chart.hs: Rendering 2D charts;
- Utf8Braille.hs: Drawing in terminal with Unicode Braille characters;
- AsciiTable.hs: Rendering of an Ascii table;
- Matrix.hs: Customizing the user interface using the Matrix theme;
- Cli.hs: Parsing command line options;
- Tui.hs: Terminal user interface.

File specs

Moex File Specification for Futures

FUTREPORT = YYYY-TRADABLESYMBOL-TLD.EXTENSION
where
 YYYY - settlement year, 4 symbols (0000-9999);
 TRADABLESYMBOL = CMY
 where
  C - code of the underlying asset, 2 symbols;
  M - settlement month (for options it also
      determines contract type - call or put),
      1 symbol
  where
   M: Jan=F, Feb=G, Mar=H, Apr=J, May=K, Jun=M,
      Jul=N, Aug=Q, Sep=U, Oct=V, Nov=X, Dec=Z;
  Y - settlement year, 1 symbol (0-9).;
 TLD - top-level domain (ru);
 EXTENSION - file extension (csv).
Example of a report name:
2012-RIZ2-ru.csv, 2022-RIZ2-ru.csv
The leading year (YYYY) is added to eliminate
confusion associated with the ten-year cycle.

Moex File Specification for Options

OPTREPORT = YYYYMMDD-FILENAME-TLD.EXTENSION
where
 YYYYMMDD - complete calendar date. For more
            information, see ISO 8601;
 FILENAME - moex file name (rts-results,
            pr-options-delta);
 TLD - top-level domain (ru);
 EXTENSION - file extension (csv).
Example of a report name:
20220103-rts-results-ru.csv,
20220103-pr-options-delta-ru.csv.



Data specs

Description of fields in CSV for Futures traded on Moex
For, like, 2022-RIZ2-ru.csv
Date|Average Weighted Price for the lot|Settlement Price|First Trade|High|Low|Last Trade|Change,%|Last Trade Volume, contracts|Number of trades|Total Volume, RUR|Total Volume, Contracts|Open Interest, RUR|Open Interest, Contracts|

Description of fields in CSV for Options traded on Moex
For, like, 20220103-rts-results-ru.csv
Contract code|Delivery|Weighted average price for the lot|Settlement Price|First trade|Max|Min|Last trade|Change|Last trade volume (contracts)|Trades num|"Trades Volume, RUR"|Trades Volume|"Open Interest, RUR"|"Open Interest, Contracts"|

Description of fields in CSV for Options traded on Moex
For, like, 20220103-pr-options-delta-ru.csv (published in Russian only)
Дата|Код опциона|Право по опциону|Последний торговый день, в который может быть заключен опционный контракт|Цена исполнения опционного контракта|Количество подлежащих исполнению опционных контрактов|Сумма подлежащих исполнению опционных контрактов|Коэффициент дельта|Теоретическая цена опциона|Волатильность|

Trade results
https://www.moex.com/en/derivatives/contracts.aspx
https://www.moex.com/ru/derivatives/contracts.aspx
The values of the “Trades Volume” field are retrieved
https://www.moex.com/en/archive/fortsmarketresults.aspx
https://www.moex.com/ru/archive/fortsmarketresults.aspx
The values of the “Волатильность” field are retrieved
https://www.moex.com/ru/forts/coefficients-values.aspx



Market Conditions (Asset Price Dynamics)

Motion Model of Asset Prices

For definiteness, consider options on futures and denote the current futures price with the symbol F. F can be understood as the current price of any underlying asset. It is assumed that asset price dynamics during the trading session is described by a continuous random process without jumps between sessions. For simplicity, the price will be described as a discrete process with a time period τ: $F0=F,F1,F2,\ldots,Fi$. The time period can be one day, one hour, one minute, etc. The time period will be expressed in fractions of a year. Since the process exists only during trading sessions, there are normally 252 trading days in a calendar year, 63 trading days in a quarter, and 21 trading days in a month. For example, if the time period is one day, then $\tau=1/252$.

A discrete-time model for prices modeled as geometric Brownian motion is given by

$$\mathbf{(Fi - Fi-1)/Fi-1 = \mu\tau + \sigma\sqrt{\tau}Z_k}$$

Where

$(Fi - Fi-1)/Fi-1$: the relative change

μ: the average rate of the price trend, expressed as a simple annual percentage

σ: volatility

$Z1,Z2,\ldots,Zi$: independent Gaussian random variables with zero mean and a standard deviation of 1 (comes from a Normal distribution)

Market Conditions (Methods of Volatility Estimation)

				WMA(100)			W100	0,1	WMA(50)			W50	0,2	WMA(20)			W20	0,3	WMA(10)			W10	0,4			
TradeDate	Settle	Ui	U²i	δ²n	δ	σ	Ū	μ	δ²n	δ	σ	Ū	μ	δ²n	δ	σ	Ū	μ	δ²n	δ	σ	Ū	μ	δ²n comp	σ comp	σ transpos
16.03.2023	76609	0,010073	0,000101	0,000107	0,010368	0,16459	0,002414	0,608387	6,85E-05	0,008278	0,131408	0,001364	0,343822	7,37E-05	0,008586	0,1363	0,002567	0,646892	5,83E-05	0,007636	0,121221	0,002635	0,664351	6,47E-05	0,126509	
17.03.2023	76853	0,003192	1,02E-05	0,000105	0,01027	0,163031	0,002418	0,609351	6,59E-05	0,00812	0,128895	0,00125	0,314916	6,66E-05	0,008158	0,129512	0,00232	0,584523	5,2E-05	0,00721	0,114461	0,002399	0,604482	6,99E-05	0,132119	
20.03.2023	77247	0,005127	2,63E-05	0,000104	0,010186	0,161704	0,002338	0,589257	6,4E-05	0,008001	0,127014	0,001254	0,316019	6,12E-05	0,007823	0,124188	0,002052	0,517164	4,9E-05	0,006998	0,111091	0,002587	0,65191	6,45E-05	0,12672	
21.03.2023	76825	-0,00546	2,98E-05	0,000102	0,010107	0,160445	0,002408	0,606847	6,22E-05	0,00789	0,125244	0,001291	0,325373	5,66E-05	0,007522	0,119404	0,002352	0,592672	4,63E-05	0,006806	0,108037	0,002247	0,566121	6,11E-05	0,123266	
22.03.2023	77175	0,004556	2,08E-05	0,0001	0,010019	0,159052	0,002319	0,584334	6,01E-05	0,007755	0,123103	0,001897	0,478041	5,16E-05	0,007181	0,113999	0,002109	0,531374	4,15E-05	0,006445	0,102318	0,00195	0,4864	5,82E-05	0,120129	
23.03.2023	75986	-0,01541	0,000237	0,000103	0,010146	0,161064	0,002139	0,539039	6,7E-05	0,008187	0,129962	0,001571	0,395947	6,75E-05	0,008217	0,130439	0,001302	0,328062	7,69E-05	0,008767	0,13917	0,000477	0,120288	5,42E-05	0,115653	
24.03.2023	77053	0,014042	0,000197	0,000105	0,01023	0,1624	0,002345	0,590846	7,22E-05	0,008494	0,13484	0,002176	0,548377	7,85E-05	0,008861	0,140656	0,001683	0,424217	0,000101	0,010028	0,159194	0,001764	0,444001	7,47E-05	0,136899	
27.03.2023	76845	-0,0027	7,29E-06	0,000103	0,010128	0,160781	0,002336	0,588772	6,99E-05	0,00836	0,132703	0,002362	0,595131	7,07E-05	0,008407	0,133452	0,000878	0,221233	8,62E-05	0,009284	0,147376	0,002662	0,670929	8,87E-05	0,149083	
28.03.2023	77125	0,003644	1,33E-05	0,000101	0,010031	0,159239	0,002267	0,571261	6,8E-05	0,008243	0,13086	0,002165	0,545625	6,42E-05	0,008014	0,127222	0,002254	0,5681	7,53E-05	0,008675	0,137715	0,002553	0,668486	7,99E-05	0,141605	
29.03.2023	77378	0,00328	1,08E-05	9,86E-05	0,009931	0,157658	0,002297	0,578834	6,61E-05	0,008128	0,129023	0,002302	0,580045	6,02E-05	0,007759	0,123166	0,002161	0,544523	6,39E-05	0,007993	0,126882	0,002034	0,51268	7,3E-05	0,135348	
30.03.2023	77445	0,000866	7,5E-07	9,64E-05	0,009821	0,155898	0,002437	0,614187	6,38E-05	0,007986	0,126772	0,002119	0,533913	5,53E-05	0,007436	0,118038	0,001875	0,472503	5,21E-05	0,00722	0,114608	0,001114	0,280654	6,67E-05	0,129273	
31.03.2023	77787	0,004416	1,95E-05	9,47E-05	0,009729	0,15445	0,002485	0,626307	6,23E-05	0,007893	0,1253	0,002376	0,598724	5,24E-05	0,007237	0,114878	0,001817	0,457997	4,56E-05	0,006753	0,107199	0,001236	0,311511	5,98E-05	0,122199	
03.04.2023																								5,59E-05	0,117848	

There are 4 types of volatility:
1. Future (“True”) Volatility (No one can get it);
2. Historical Volatility (It is calculated exactually);
3. Forecast Volatility (It is guessed or predicted);
4. Implied Volatility (one method is using options' market price).

Historical volatility is determined by the prices of the underlying asset.

In order to get an estimate of the μ, σ parameters for a discrete set of prices of the underlying asset F0=F,F1,F2,…,Fi, it is necessary to determine the relative price changes of the period τ

Settle: Settlement Price

Ui: The Relative Change: $Ui=(Fi - Fi-1)/Fi-1$, for period τ

U^2i: The Square of Relative Change

to calculate the average speed of the trend

Ū: The average speed of the trend: $Ū=1/m*\sum Un-i$

then calculate the volatility estimate of the period τ

$δ^2n$: Fishburne Weighted Variance: $δ^2n=\sum αi*U^2n-i$, $\sum αi=1$

δ: Weighted Evaluation of Volatility: $δ=\sqrt{δ^2n}$, for period τ

δ is the standard deviation of random variables U1,U2,…,Ui.

Estimates of the coefficients μ, σ are obtained by normalization.

μ: The average rate of the price trend, expressed as a simple annual percentage: $μ=Ū/τ$

σ: Evaluation of δ: $σ=δ/\sqrt{τ}$, $τ=1/252$

Window Principle

Count Window: Count Window groups data streams according to the number of elements

Sliding Count Window: Calculate the sum of the most recent 100 elements every 10 elements

Fishburne formula

$αi=2*(n-i+1)/(n*(n+1))$

HV Interface

HOLS	WSAT	Param	Value	Weight
02.01.2023		Unit	1	
23.02.2023		FNetWH	0,6042	
08.03.2023		FNonWH	0,3958	
01.05.2023		NetWD	254	
09.05.2023		NetWHWD	153,4668	1
12.06.2023		NonWHWD	100,5332	1
		NonWD	0	0
		CompLOT	254	
		ExpDate	19.04.2023	
		NetDTE	12	
		NetWHDTE	7,2504	1
		NonWHDTE	4,7496	1
		FreeDTE	0	0
		CompDTE	12	

Directory	C:\Users\0mega3\Desktop\esys\data\buf\autohv\SBRF\
JoinMode	AM
JoinDepth	10
Estimator	Vyz
Frequency	1
LOT	254
DTE	18
BegDate	24.03.2023
EndDate	24.03.2023

Settle	Open	High	Low	Close						
TradeDate	VarDTE	VolaDTE	Var100	Vola100	Var50	Vola50	Var20	Vola20	Var10	Vola10

Trading Schedule

Main trading session
09:00-10:00=60M;
10:00-14:00=240M;
Intraday clearing session
14:00-14:05=5M;
Main trading session
14:05-18:50=285M;
Evening clearing session
18:50-19:05=15M
Evening trading session
19:05-23:50=285M;

Unit: Unit Converter

From 1D to H (Unit=1 to Unit=24);
From 1D to M (Unit=1 to Unit=1440);
From 1D to S (Unit=1 to Unit=86400)

FNetWH: Fraction of Net Working Hours

Net trading time (NetTT)=870M;
FNetWH=NetTT/1440=0.6042

FNonWH: Fraction of Non-Working Hours

Non-trading time (NonTT)=570M;
FnonWH=NonTT/1440=0.3958

NetWD: Net Workdays

The number of whole working days between start_date and end_date. Working days exclude weekends and any dates identified in holidays (HOLS). It is also adjusted for working Saturdays (WSAT).

NetWHWD: Net Working Hours of Workdays

NetWHWD=NetWD*FNetWH*Weight

NonWHWD: Non-Working Hours of Workdays

NonWHWD=NetWD*FNonWH*Weight

NonWD: Non-Working Days

The length of the Gregorian year (LGY) consists of 365 days (common year), with a leap day being added to February in the leap years.

NonWD=LGY*Unit-(NetWHWD+NonWHWD)

CompLOT: Composite Length of Time

CompLOT=NetWHWD+NonWHWD+NonWD

ExpDate: Option Expiration

NetDTE: Net Days to Expiration

NetWHDTE: Net Working Hours of Days to Exp

NetWHDTE=NetDTE*FNetWH*Weight

NonWHDTE: Non-Working Hours of Days to Exp

NonWHDTE=NetDTE*FNonWH*Weight

FreeDTE: Free Days to Expiration

The number of time intervals (NTI) between two specified dates.

FreeDTE=NTI*Unit-(NetWHDTE+NonWHDTE)

CompDTE: Composite Days to Expiration

CompDTE=NetWHDTE+NonWHDTE+FreeDTE

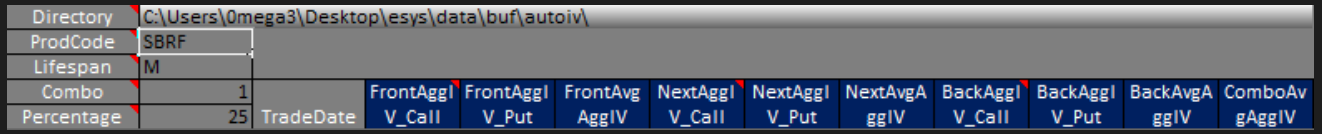
HV Interface (continued)

Directory: The Path to Market Data
JoinMode: The mode of joining between the selected fields from the prices of the expired and continuing contracts
Price gaps on contract rollover are adjusted to avoid wrong decisions.
The fields used to calculate the adjustment are set by default (Settlement, OHLC prices).
JoinMode parameters: AM (Arithmetic mean (by default)), GM (Geometric mean)
JoinDepth: The depth of joining of the expired and continuing contracts
It is determined by the number of trading days before expiry.
Rollover trigger:
If JoinDepth=1 then Rollover strictly on the exp day of the contract;
If JoinDepth=0 then Rollover strictly on the day after the exp day of the contract;
If JoinDepth>1 then Rollover a number of days before the exp day of the contract
Estimator: Computational Methods of Historical Volatility (HV)
Abbr: Prices Taken(PT), Handle Drift(HD), Handle Overnight Jumps(HOJ), Efficiency(E)
Estimator parameters:
Vcc (Close-to-Close (PT: C; HD: No; HOJ: No; E(max): 1))
Vp (Parkinson (PT: HL; HD: No; HOJ: No; E(max): 5.2)),
Vgk (Garman-Klass (PT: OHLC; HD: No; HOJ: No; E(max): 7.4))
Vrs (Rogers-Satchell (PT: OHLC; HD: Yes; HOJ: No; E(max): 8))
Vgkyz (Garman-Klass Yang-Zhang ext. (PT: OHLC; HD: No; HOJ: Yes; E(max): 8))
Vyz (Yang-Zhang (PT: OHLC; HD: Yes; HOJ: Yes; E(max): 14))
If the “Estimator” field is empty, then prices are displayed.

Frequency: Frequency of Measurement (e.g. minutely/hourly/daily/weekly/monthly)
LOT: Length of Time (e.g. number of minutes/hours/days/weeks/months)
DTE: Days to Expiration
If the “DTE” field is empty, then it is calculated:
Var100: 100-day variance;
Vola100: 100-day volatility;
Var50: 50-day variance;
Vola50: 50-day volatility;
Var20: 20-day variance;
Vola20: 20-day volatility;
Var10: 10-day variance;
Vola10: 10-day volatility.
BegDate: Begin Date
If the “BegDate” and “EndDate” fields are empty, then the workspace is cleared.
EndDate: End Date
VarDTE: DTE-day variance
VolaDTE: DTE-day volatility
Settle: Settlement price



IV Interface

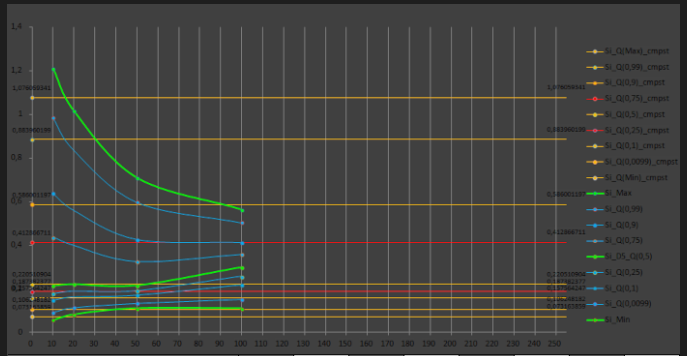


Directory	C:\Users\0mega3\Desktop\esys\data\buf\autoiv\										
ProdCode	SBRF										
Lifespan	M										
Combo	1		FrontAggIV_Call	FrontAggIV_Put	FrontAvgAggIV	NextAggIV_Call	NextAggIV_Put	NextAvgAggIV	BackAggIV_Call	BackAggIV_Put	BackAvgAggIV	ComboAvgAggIV
Percentage	25	TradeDate										

Directory: The Path to Market Data
ProdCode: The underlying asset code also known as Product Code
If the “ProdCode” field is empty, then the workspace is cleared.
Lifespan: Options Lifespan
All options have a limited useful lifespan (Weekly (W), Monthly (M), Quarterly (Q)) and every option contract is defined by an expiration month.
Combo: Options Series Combinator
If the “Combo” field is empty, then it is set to 1 by default;
If the “Combo” field is set to 1, there is no combo;
If the “Combo” field is set to 2, then Front IV and Next IV are combined;
If the “Combo” field is set to 3, then Front IV, Next IV and Back IV are combined.
Percentage: The percentage (pct) is strike range: +/- pct% of the ATM (At-The-Money) strike
If the “Percentage” field is empty, then it is set to 0% by default.
Percentage takes into account zero volumes only.
The aggregated (agg) IV (the Fishburne formula is used) consists of: ATM IV, Weighted IV by non-zero volume, IV with zero volumes above the CS (central strike), IV with zero volumes below CS.

FrontAggIV_Call: The front agg IV of the call option.
FrontAggIV_Put: The front agg IV of the put option.
FrontAvgAggIV: The average between the front agg IV of the call option and the front agg IV of the put option.
A front month is the nearest expiration date for a futures or options contract.
NextAggIV_Call,
NextAggIV_Put,
NexttAvgAggIV:
A next month is the expiration date for a futures or options contract in the following month.
BackAggIV_Call,
BackAggIV_Put,
FrontAvgAggIV:
Back-month contracts have later expiration dates than front-month contracts and are also called for-month contracts.

VC Interface



	Si			Depth_HV	2	Depth_IV	2	Lifespan	M	Combo	1
HalfPipe	Lip	Vert	Transition			Flat		Transition		Vert	Lip
Decile	Min	D1	D2	D3	D4	D5	D6	D7	D8	D9	Max
IV	0,125914	0,135996	0,143214	0,149534	0,153673	0,157814	0,160175	0,164537	0,171673	0,181547	0,212387
HV_10d	0,057529	0,099291	0,122453	0,143379	0,173415	0,21423	0,270748	0,34139	0,427024	0,584222	1,21054
HV_20d	0,083371	0,112328	0,13279	0,149606	0,174775	0,22117	0,268968	0,394875	0,476236	0,61211	1,014549
HV_50d	0,110414	0,132609	0,154661	0,17943	0,198431	0,215696	0,323112	0,469964	0,514436	0,604575	0,708156
HV_100d	0,110649	0,144022	0,17016	0,185765	0,207099	0,299657	0,411357	0,505634	0,519754	0,533109	0,562011
Quantile(P)	0,0013	0,0099	0,1	0,25		0,5		0,75	0,9	0,99	0,9987
Z-score	-3	-2,3264	-1,2816	-0,6745		0		0,6745	1,2816	2,3264	3
Q(P)IV		0,133076	0,144186	0,150642				0,170084	0,181127	0,200134	
Q(P)HV_10d		0,092714	0,147287	0,178999				0,438234	0,639854	0,986835	
Q(P)HV_20d		0,114312	0,162302	0,190188				0,399548	0,560101	0,836409	
Q(P)HV_50d		0,134053	0,170719	0,192025				0,326417	0,426075	0,597582	
Q(P)HV_100d		0,153088	0,218913	0,257162				0,358643	0,411735	0,503104	
ΣW=1	0,5	0,4	0,06	0,04							
Q(P)HV_cmpst	0,073164	0,106248	0,157564	0,187382		0,220511		0,412867	0,586001	0,88396	1,076059

DepthHV: Depth of HV data
Expressed in years.
DepthIV: Depth of IV data
Expressed in years.
Lifespan: Options Lifespan
Combo: Options Series Combinator

HalfPipe: Decile encapsulation zones
Lip; Vert; Transition; Flat; Transition; Vert; Lip
Min; D1 ; D2,D3,D4 ; D5 ; D6,D7,D8 ; D9 ; Max
Decile: Min, D1, D2, D3, D4, D5, D6, D7, D8, D9, Max
A decile is a quantitative method of splitting up a set of ranked data into 10 equally large subsections.[3]
Quantile(P): Inverse value of the standard normal distribution function
In statistics and probability, quantiles are cut points dividing the range of a probability distribution into continuous intervals with equal probabilities, or dividing the observations in a sample in the same way. There is one fewer quantile than the number of groups created.[4]
Z-score: The z-score of a normal distribution
Z-score, or standard score, is the number of standard deviations a given data point lies above or below the mean. The mean is the average of all values in a group, added together, and then divided by the total number of items in the group.[5]
StdDev: −3σ; −2σ; −1σ; 0; +1σ; +2σ; +3σ
Z-score: -3.0; -2.0; -1.0; 0; +1.0; +2.0; +3.0
Quantile: 0.0013; 0.0099; 0.1; 0.25;0.5; 0.75; 0.9; 0.99;0.9987
Z-score: -3;-2.3264;-1.2816;-0.6745; 0;0.6745;1.2816;2.3264;3
The upper limit of the VC (Max column)
The middle line of the VC (D5 column)
The lower limit of the VC (Min column)
General formula for calculating quantile values by Z-score:
D5Value+(D5Value-MinValue)/3*(-Zscore)
D5Value+(MaxValue-D5Value)/3*(+Zscore)

Conslusion

Why options?

Directional Trading

Directional trading refers to strategies based on the investor's view of the future course of something: either the overall financial market or a particular security. Their assessment of the direction will be the sole determining factor in whether the investor decides to sell or buy.[6]

- Directional trading with strictly fixed risk (purchase of call and put options);
- Directional trading, like trading an underlying asset, using synthetic futures (buying and selling call and put options).

Hedging

The best way to understand hedging is to think of it as a form of insurance. When people decide to hedge, they are insuring themselves against a negative event's impact on their finances. This doesn't prevent all negative events from happening. However, if a negative event does happen and you're properly hedged, the impact of the event is reduced.[7]

Market Making

Making a market is an action whereby a dealer or market maker stands by, ready to make a two-sided quote. This quote indicates they are willing and able to either buy or sell a particular security at the quoted bid and ask price.[8]

Trading Volatility

- Trading Implied Volatility (Kappa/Vega risk);
- Replication of options (Delta-Gamma risk)



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